Deepfake Recognition Project Proposal

CNN Based Detection of Manipulated Facial Images and Videos

Selected track	Fraud and identity protection	**Team number**	Team 6
Team members	Flemming Grabowski & Payel Sarkar	**Submission date**	December 6, 2026
Repository	https://github.com/payelsarkarr/IS665-Team6-Deepfake-Recognition	**Dataset**	FaceForensics++

1. Problem Definition and Security Significance

Deepfakes are fake or edited images and videos created with artificial intelligence. They can make a person appear to say or do something that never happened. Criminals may use deepfakes to impersonate family members or company leaders and trick people into sending money. They may also place someone's face into a harmful image, use it for blackmail, or upload it to an unsafe website. The FBI has warned that AI-generated media is being used for financial fraud, identity impersonation, and extortion (FBI Internet Crime Complaint Center, 2024).

Team 6 proposes a convolutional neural network, or CNN, that will check facial images and short videos and classify them as likely real or fake. Individuals and security teams could use the result as an early warning. It may help someone stop before sending money, sharing personal information, or responding to a threat. The model will support human review. It will not be treated as legal proof or as a guarantee that an image is real.

Project question: Can a CNN identify deepfake facial images and videos while keeping missed fakes and false warnings reasonably low?

2. Project Scope and Success Criteria

The prototype will accept a facial image or short video. For a video, it will select a few frames, find the main face, and check each frame. It will combine the results into one final score. The project will focus only on visual deepfakes. It will not examine cloned voices, identify the person, search websites, remove online content, or determine who created the deepfake.

Success criterion	Target or decision rule	Security reason
Detect manipulated media	Deepfake recall of at least 0.80 on the held-out test set	Missing a fake may expose a user to fraud, impersonation, or blackmail.
Limit false accusations	False-positive rate at or below 0.20	A real image should not be labeled fake without sufficient evidence.
Explain the warning	Show frame scores and a simple face heatmap	Users should see what influenced the warning.

3. Proposed Dataset

We will use FaceForensics++, a research dataset made from 1,000 original videos and four types of face manipulation. The complete dataset contains more than 1.8 million manipulated images (Rössler et al., 2019). The full collection is too large for a two-person project, so we will use an approved sample of about 200 real and 200 fake videos. We will take about 10 frames from each video, giving us roughly 4,000 face images. We will report the final number after checking the files.

Field	Planned value
Unit of analysis	Training: one cropped facial frame. Demonstration: one still image or short video.
Target variable	Binary label: real or manipulated.
Expected balance	Approximately 50 percent real and 50 percent manipulated in the selected subset.
Access and rights	Access requires an application and compliance with the FaceForensics Terms of Use. Files will not be redistributed.
Quality concerns	Blurry or compressed images, failed face detection, repeated frames, and few examples of newer deepfakes.
Split strategy	Grouped 70/15/15 train, validation, and test split by source video before frame extraction.

We will divide the videos before taking frames from them. All frames from the same original video will stay in one group. This prevents the model from seeing nearly identical images during both training and testing. We will not collect new images from classmates or members of the public.

4. Proposed Technical Approach

We will check the labels, remove unreadable or repeated images, select frames from each video, crop the main face, and resize it to 224 by 224 pixels. We will compare the number of real and fake images and check image quality. During training, we may slightly flip, resize, brighten, or compress images so the model learns from realistic changes.

Stage	Planned method	Evidence to produce
Baseline	Small CNN trained on the selected frames	Learning curves and baseline test metrics
Main model	MobileNetV2 with transfer learning	Model comparison and training-time record
Evaluation	Precision, recall, F1, ROC-AUC, confusion matrix, and video-level voting	Test results and false-positive/false-negative review
Robustness	Repeat testing after JPEG compression and resizing	Performance change under common online processing
Prototype	Simple Streamlit page with selected frames, confidence score, and face heatmap	Working demonstration and user instructions

We will use Python, Keras, OpenCV, scikit-learn, and Streamlit. MobileNetV2 is already trained to recognize useful image patterns, so adapting it should require less time and computing power than building a large model from the beginning.

5. Risks Ethics and Responsible AI

A detection score can reduce risk only when it is interpreted carefully. A false negative could allow fraudulent content to appear trustworthy, while a false positive could wrongly damage a person's reputation. The interface will therefore use language such as likely manipulated and needs review. It will not label a person a criminal or automatically publish a result.

Risk	Likelihood and impact	Planned mitigation or test
New deepfake methods	High: new tools may create different visual signs.	Test one unseen manipulation type when possible and clearly state the model's limits.
False positives and negatives	High: either error can cause financial or personal harm.	Report both error types, tune the threshold on validation data, and require human review.
Bias	Medium to high: results may differ by skin tone, age, lighting, or image quality.	Compare available groups and clearly report missing demographic information.
Privacy and harmful media	High: facial images and abusive content are sensitive.	Use only approved research data, restrict access, avoid collecting victim images, and delete uploads after analysis.
Edited or blurry files	Medium: cropping, blur, or compression may hide signs of a deepfake.	Test resized and compressed images and avoid claiming that the model catches every fake.
Difficult model decisions	Medium: CNN results may be hard to explain.	Show a face heatmap and confidence score with every result.
Computing and schedule	Medium: video processing and training may be slow.	Use a smaller dataset, transfer learning, and saved face crops.

6. Work Plan and Team Roles

The schedule assumes six weeks of work and can be changed to match the course deadlines. Both members will review the code, results, and final report.

Milestone	Owner	Target date	Planned output
Access approval and repository setup	All members	Sep 20, 2026	Approved access, README, and task board
Dataset audit and frame preparation	Payel	Sep 27, 2026	Quality notes and prepared face images
Baseline CNN and interim report 1	Flemming	Oct 4, 2026	Notebook and baseline results
Main model and error review	Flemming and Payel	Oct 10, 2026	Model comparison and error examples
Testing, ethics, and interim report 2	Payel	Nov 5, 2026	Compression test and risk review
Prototype and final report	Flemming	Dec 3, 2026	Simple demonstration and report draft
Final review and presentation	Flemming and Payel	Dec 6, 2026	Final PDF, slides, video, and repository

Preliminary responsibilities:

Payel Sarkar will lead data access, image preparation, and testing.

Flemming Grabowski will lead CNN training, evaluation, and the demonstration page.

Both members will review the code, ethics, report, and presentation so that each person understands the complete project.

References

Chollet, F. (2017). Xception: Deep learning with depthwise separable convolutions. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (pp. 1251-1258). https://doi.org/10.1109/CVPR.2017.195

FBI Internet Crime Complaint Center. (2024, December 3). Criminals use generative artificial intelligence to facilitate financial fraud. https://www.ic3.gov/PSA/2024/PSA241203

Rössler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., & Nießner, M. (2019). FaceForensics++: Learning to detect manipulated facial images. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 1-11). https://doi.org/10.1109/ICCV.2019.00009

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In Proceedings of the IEEE International Conference on Computer Vision (pp. 618-626). https://doi.org/10.1109/ICCV.2017.74

Technical University of Munich. (2019). FaceForensics++ dataset and access instructions. https://github.com/ondyari/FaceForensics

Dataset and Repository Links

Dataset access: FaceForensics++ official repository

Team repository: *https://github.com/payelsarkarr/IS665-Team6-Deepfake-Recognition*

Appendix A Team Contribution Statement

The following assignments describe the planned division of work at the proposal stage. The team will update this table with completed work and repository evidence before the final submission.

Team member	Responsibilities	Expected evidence
Payel Sarkar	Dataset access, frame extraction, face detection, data checks, and robustness testing	Preprocessing notebook, quality notes, test results, and repository commits
Flemming Grabowski	CNN training, model evaluation, Streamlit demonstration, heatmap, and documentation	Training notebook, saved model, application, report sections, and repository commits

The team will coordinate through a shared repository, weekly meetings, and peer review of notebooks and written sections. Workload concerns will be raised during weekly meetings and tasks will be reassigned when needed.

AI Use Disclosure

Tool	How it was used	What the team will verify or change
ChatGPT	Topic brainstorming, proposal organization, and wording support	The team will verify every technical claim and citation, revise the text in its own voice, confirm the dates and scope, and remain responsible for the submitted work.
Claude	Proposal review, organization suggestions, and wording support	The team will verify every technical claim and citation, revise the text in its own voice, confirm the dates and scope, and remain responsible for the submitted work.

Submission Checklist

Status	Requirement
Complete	Confirm the submission date.
Complete	Create the repository and add the final link.
Pending	Confirm FaceForensics++ access and the permitted subset.
Complete	Proposal body is three pages; references and appendix are separate.
Complete	Sample prompts and instructional text have been removed.
Pending	Every team member reviews the final submission and filename.

Rubric Self Check

Criterion	Points	Evidence
Problem definition and significance	2	Section 1, page 1
Dataset and scope	2	Sections 2 and 3, pages 1-2
Technical approach and feasibility	3	Section 4, page 2
Risks and Responsible AI planning	2	Section 5, page 3
Work plan, clarity, and completeness	1	Section 6, page 3; Appendix A